

# CUTTLESWISH

HELPING

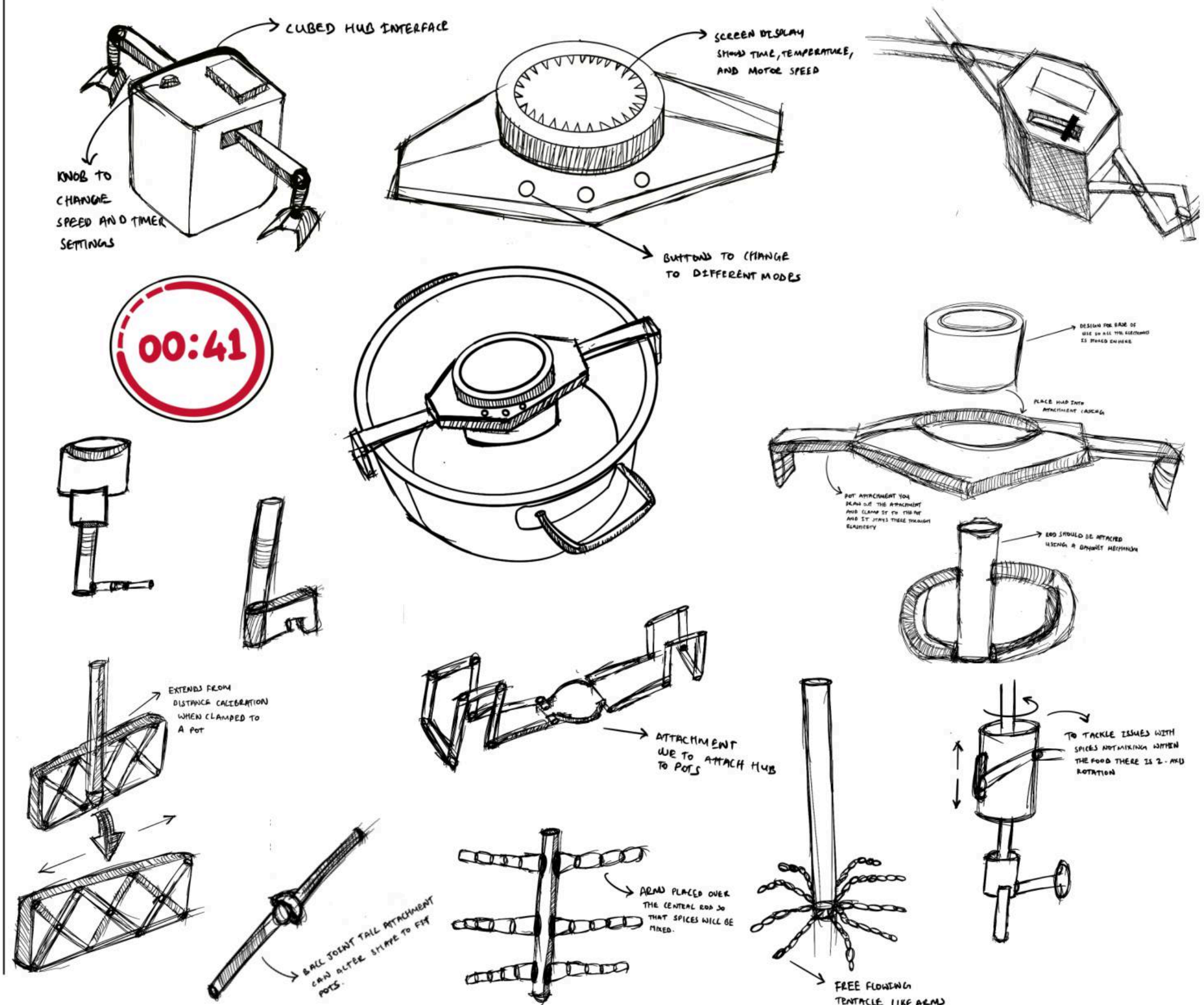
ELDERLY

COOKS





This paired visual feedback of an LED ring provided a much more elegant and effective interface for our target market to operate.







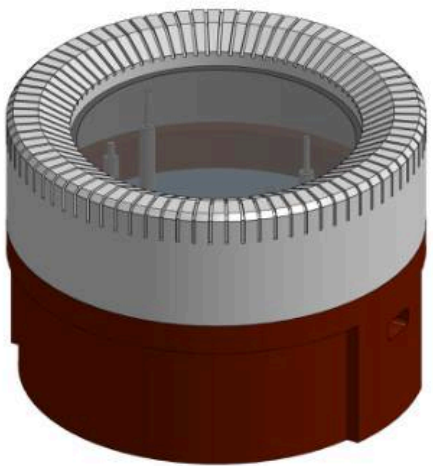
## CUTTLESW!SH

Automatic Stirrer

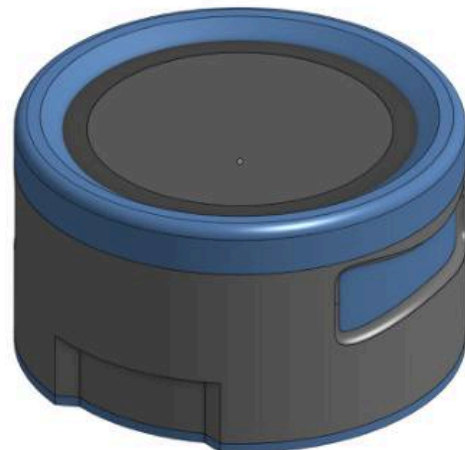
- 1 +

ADD TO CART

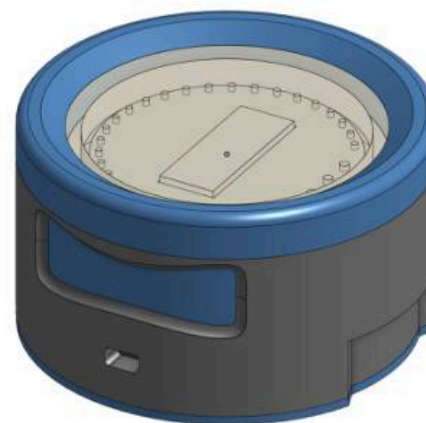
The final design was arrived to through a long development process and concluded in the cute, zoomorphic appearance that was made to reflect the look of a cuttlefish. The design came about from version 4, where we experimented with having “tentacle-like” attachments and arms to give more flexibility and customisability to the height and shape of the pot and arm it could fit in. This was the developed from physical joints, to bendable silicone and flexible support arms to reduce the required effort from cleaning. UI design also shifted throughout from touchscreens, to segmented displays and finally LED rings and buttons.



Version 1  
KitchenAid



Version 2  
JosephJoseph



Version 3  
JosephJoseph



Version 4  
KitchenAid

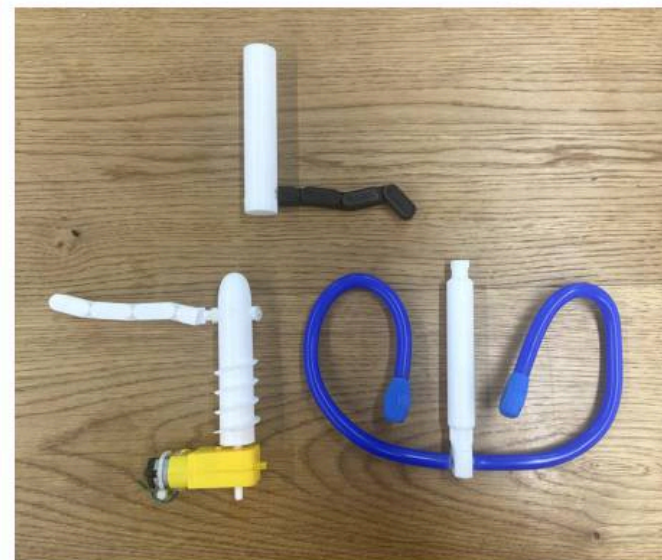
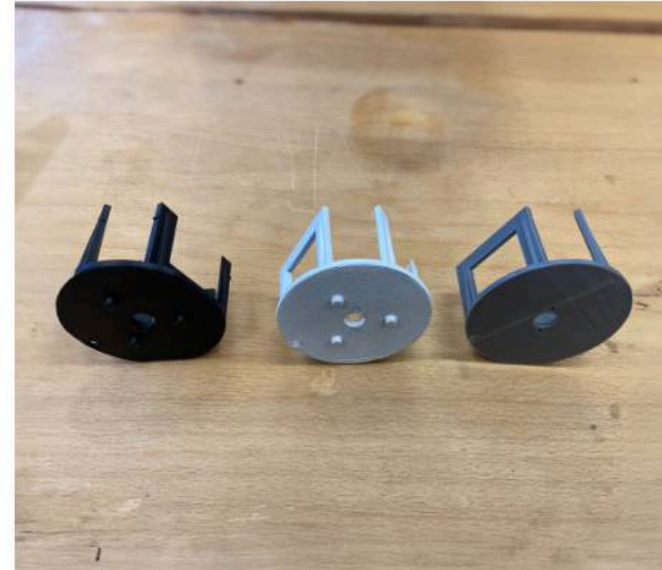


Version 5  
PELEGDESIGN



# CONCEPT DEVELOPMENT

## DESIGN ISSUES



## Issues & Solutions

### Mechanical Issues

The main mechanical issues came about when focusing on the support mechanism. Firstly, there were issues with the legs bending unevenly, which were fixed by introducing a detent mechanism in the joint.

Then, there were issues with ensuring the mechanism would attach to the pot and grip it. Testing with TPU models allowed us to find that although it would grip well enough, having the hole arm made of TPU would make it bend too much, leading to us making just the end out of TPU.

### Electronic Issues

Electronics plagued us throughout this project. Between taking long to assemble and build, and then soldered circuits continuously failing, testing and development took very long. The final model ended up having to be assembled using pin wires and could not be soldered as originally planned.

### Heat, water and food safety

We had to keep in mind throughout the design process how we would protect the electronics from the heat, steam and even splashes of food, whilst making sure it was still food safe. This was done through smart material choice and having multiple layers to protect the electronics as much as possible from external factors.



# COLOURS, MATERIALS & FINISHES

## SEMIOTIC ANALYSIS

The Cuttlesw!sh embodies a thoughtful selection of colours, materials, and finishes, ensuring not only functionality but also aesthetic appeal. The chosen materials contribute to the product's longevity and performance, while the smooth finishes add a touch of sophistication, making the Cuttlesw!sh a delightful addition to any kitchen. With the main body being kept white, the blue highlights can be changed to whatever colour the user desires.

### ABS

This is a versatile material which is both temperature resistant and, more importantly, food-safe. Its durable nature and ability to be injection moulded further adds to our desire to use it.

### Silicone

Silicone has been used in multiple aspects of the stirrer for its mouldable and food-safe properties. It covers the stirrer, ensuring hygiene and safety, seals the support arm joints to prevent food and dirt ingress, provides a grip on the support arms for better adherence to the pot, and protects the electronics on the top by sealing them securely.

### Nylon

The gears were designed to be made of nylon due to its self-lubricating properties, eliminating the need for additional grease. Additionally, nylon's durability ensures long-lasting performance in the device.

### Copper

Copper was incorporated inside the mouldable stirrer because it provides the necessary flexibility to bend while retaining its shape during cooking.



The Cuttlesw!sh™ has been designed to ensure a simple yet highly functional user experience. Every element, from the intuitive controls to the adaptable components, is crafted to enhance usability. These features detailed below illustrate how the device meets the needs of users to make cooking accessible.

### On/Off Button

The On/Off button activates the device and plays a brief LED animation accompanied by a tune, enhancing the user experience with visual and auditory feedback.

### Wing Handles

The wing handles are ergonomically designed to allow users to easily lift the main hub off the pot, providing a secure and comfortable grip.

### Mode & Start Button

Pressing this button cycles through the three available modes, while holding it starts the Cuttlesw!sh only when a timer has been set.

### LED Ring

The LED ring serves as the primary output display, indicating the timer, temperature, and motor speed through different colours.

### Flexible Arms

These flexible arms can clamp onto various surfaces, ensuring a secure attachment and stability during operation, adapting to different pot shapes and sizes.

### Mouldable Stirrer

The mouldable stirrer allows users to customise the stirring attachment to fit the specific size and function required.





The assembly of the Cuttleswish™ is designed to be straightforward and efficient, with key components that ensure robust functionality and ease of use. Below, we outline the critical features and methods used in the product's assembly.

### 3 Key Sections

The product is made up of 3 key components, the main hub which houses all the electronics, the main shell which protects the hub and holds it above the pot, and the attachment to stir the food.

### Snap & Press Fittings

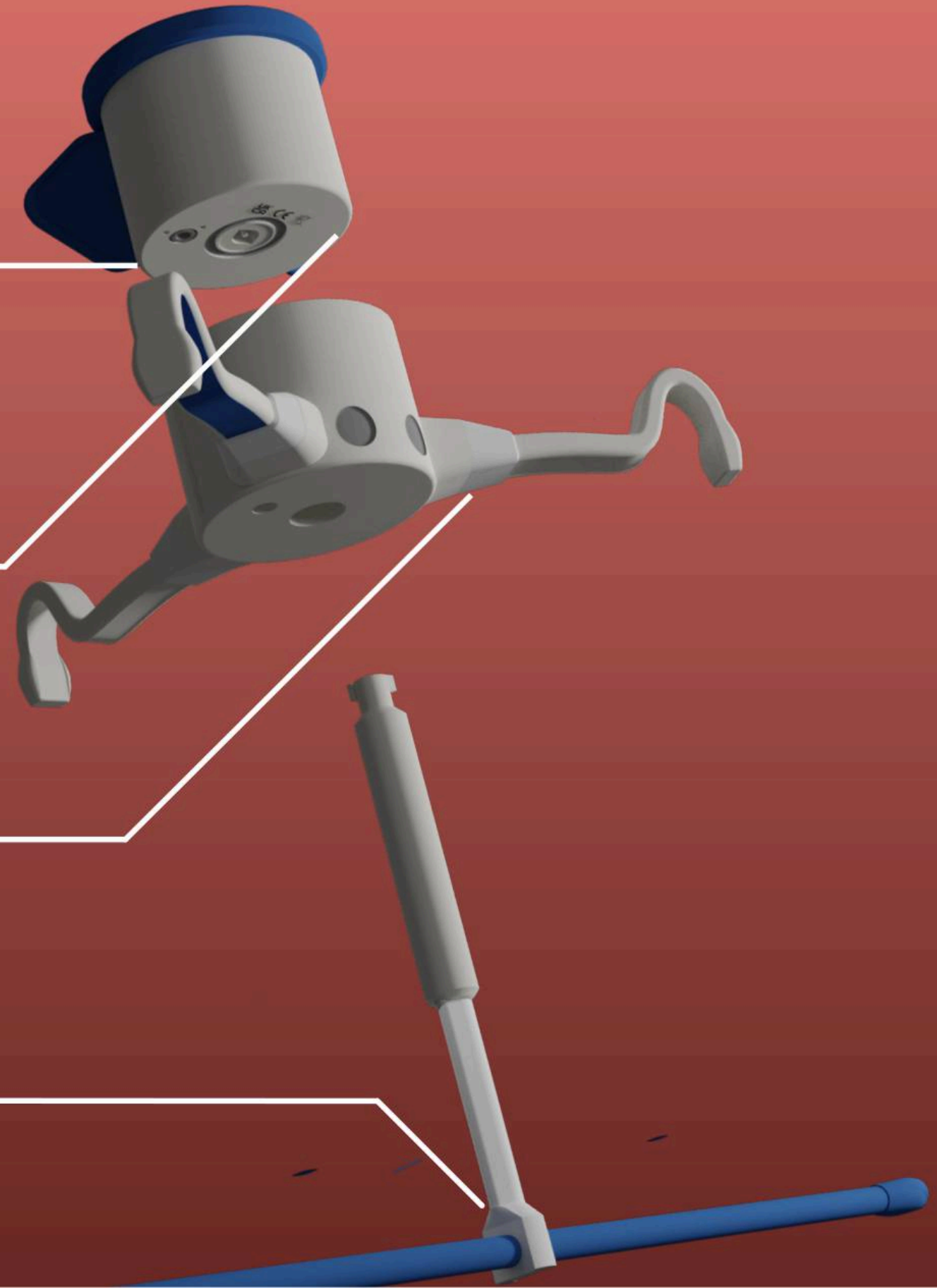
The majority of the components have been designed to be assembled by snap and press fittings allowing for us to not use more permanent attachment methods.

### Screws

Some components were made to be screwed on, such as the motors and start of arms.

### Provisioning

Three key ideas allow the different components to fit together: Clearance fits, Interference fits and Allowance & Enveloping.



## Kitchenaid

Known for its stylish and sleek finish, Kitchenaid produces high-quality kitchen appliances that are both functional and visually appealing. [1]



Artisan Mixer Tilt-head 4.8L

### Core Values

**Quality**  
Emphasis on producing durable, high-quality products.



**Innovation**  
Continual improvement and innovation in kitchen solutions.



**Community**  
Commitment to supporting and building community connections.



**Heritage**  
Creating a strong legacy of craftsmanship and design.



Est 1919  
Market Value of \$12.4 billion, 2024 [2]

## JosephJoseph

Specialising in easy-to-use, minimalist kitchen tools with a splash of colour, catering to practical and visually engaging needs. [3]



Helix Garlic Press

### Core Values

**Intentiveness**  
Creating products that solve everyday problems in original ways.



**Determination**  
Persistence in bringing daring and distinctive products to market.



**Independence**  
Following their instincts to innovate in the homeware industry.



**Collaboration**  
Emphasis on teamwork to ensure functionality and aesthetic appeal.



Est 2003  
Market Value of \$54 million, 2024 [4]

## Ototo

Renowned for its creative, quirky kitchen gadgets that blend functionality with a playful, light-hearted aesthetic. [5]



Shelly Measuring Cups

### Core Values

**Playfulness**  
Incorporating fun and humor into everyday kitchen items.



**Functionality**  
Ensuring that products are not just whimsical but also practical.



**Innovation**  
Continuously exploring new ideas and pushing design boundaries.



**Accessibility**  
Creating user-friendly products that are accessible to all customers.



Est 2004  
Market Value of N/A, 2024



## PelegDes!gn

Shahar Peleg created Peleg Design in 2004 and it is well known for its playful and practical kitchen tools. They make tedious tasks enjoyable with whimsical designs and clever product names blending putting the fun in functionality.



Wine Bottle Holder



Soap Opera



Wine Bottle Holder



Soap Opera

Peleg Design stands apart from companies like Kitchenaid, Joseph Joseph, and Ototo because of its simple, elegant, yet witty design language. Peleg's minimalistic concepts, in contrast to Ototo's: humorous yet less refined. This playful aesthetic broadens our target market for Cuttleswish to be a product for both the elderly and younger generation. [6]

## Brand Iceberg

### Humour

Shahar Peleg infuses wit into all products from their design down to the name.

### Functionality

"Our products are not just decorative; they are meant to be used every day."

### User Engagement

"We love it when people say, 'Wow, this is brilliant!'"

They make products to directly align with consumer needs.

### Attention to Detail

Every product is design to perfection from physical product to branding

### Visual Identity

Their formal design language can be seen in their typeface but is contrasted by their playful logo as seen with the inverted "i".

### Aesthetics

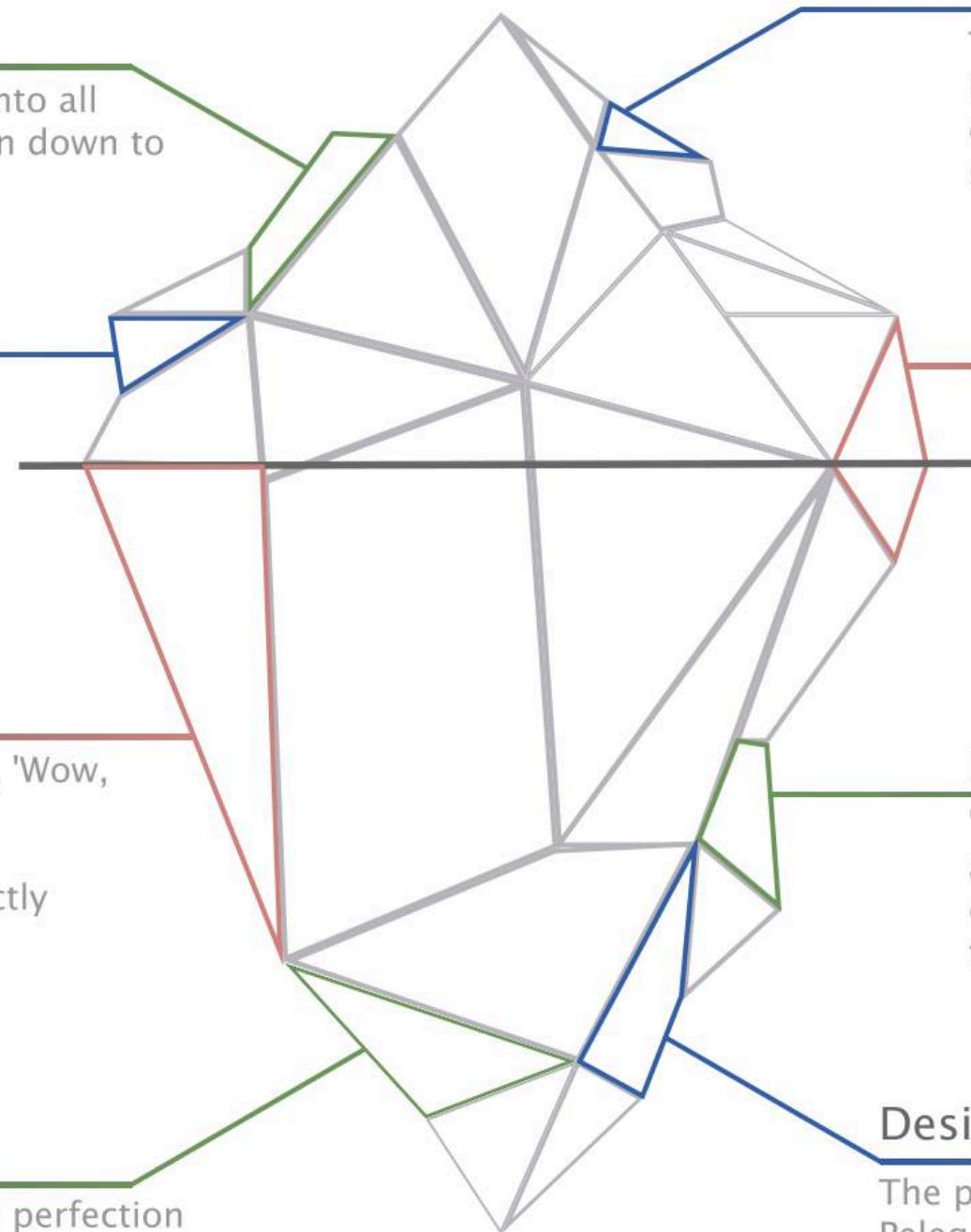
"Known for their whimsical designs and pun-based names... (Shahar) invites the viewer to take a closer look, revealing a fresh view of everyday products."

### Innovation

"Transforming what was once an average, been-there-done-that concept into an innovative and truly exciting experience."

### Design Inspiration

The primary source of inspiration for Peleg Design's designs is zoomorphism, which gives their creations vitality.

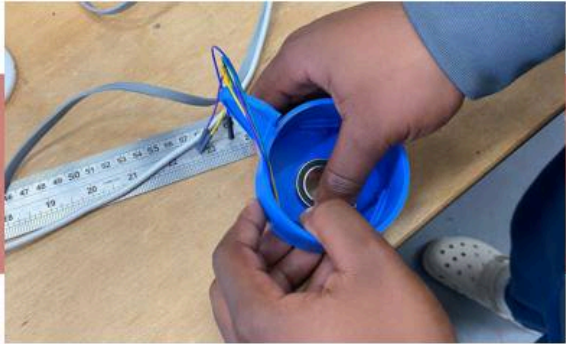




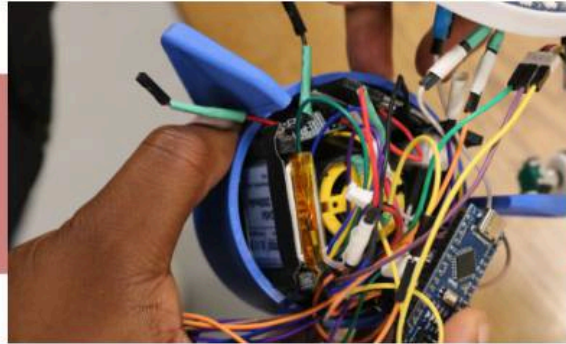
# PRODUCT ASSEMBLY PROCESS

TIMINGS & COST ESTIMATE

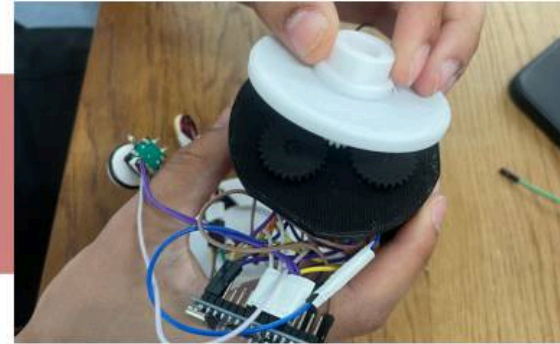
## 01 Inner Shell



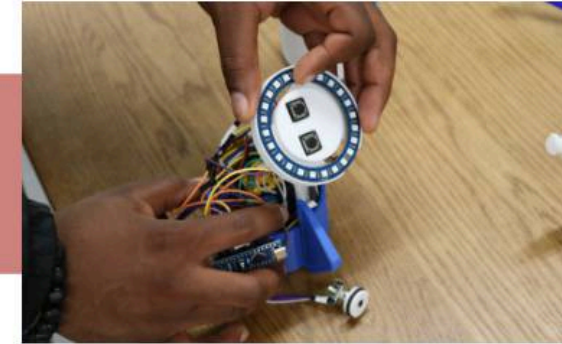
20sec – Press fitting the bearing and IR sensor in place.



15mins – Attaching the electronics to the main base.



1min – Inserting the gears.



5mins – Assembling everything in the main shell.

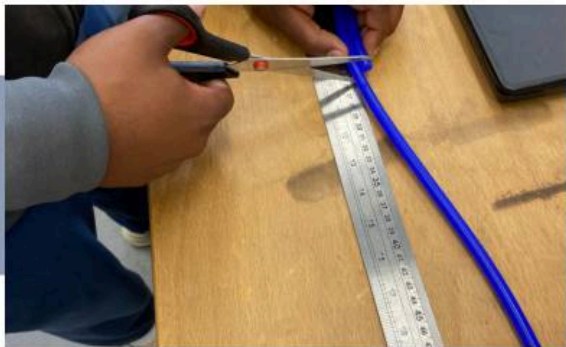


2mins – Attaching the silicone cover and top knob.

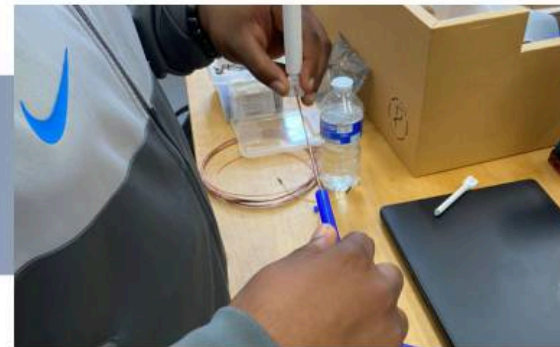
## 02 Attachment



20 sec – Snap fitting the bottom rod to the top.



3 mins – Measuring and cutting the silicone & copper.



20 sec – Inserting the copper and silicone.



5 mins – Glueing everything to the bottom attachment.



5 mins – Glueing the end of the tube stopper.

## 03 Outer Shell



1 min – Snap fitting the main arm to the outer shell.



5 mins – Glueing the TPU ends to the support arms.



10 sec – Inserting the inner shell to the main one.



20 sec – Putting the attachment in.



5 sec – Place it on the desired pot.

Total Time: 43 minutes 40 seconds

The actual total assembly time will be significantly reduced in a manufacturing setting due to the use of automated processes and machinery.

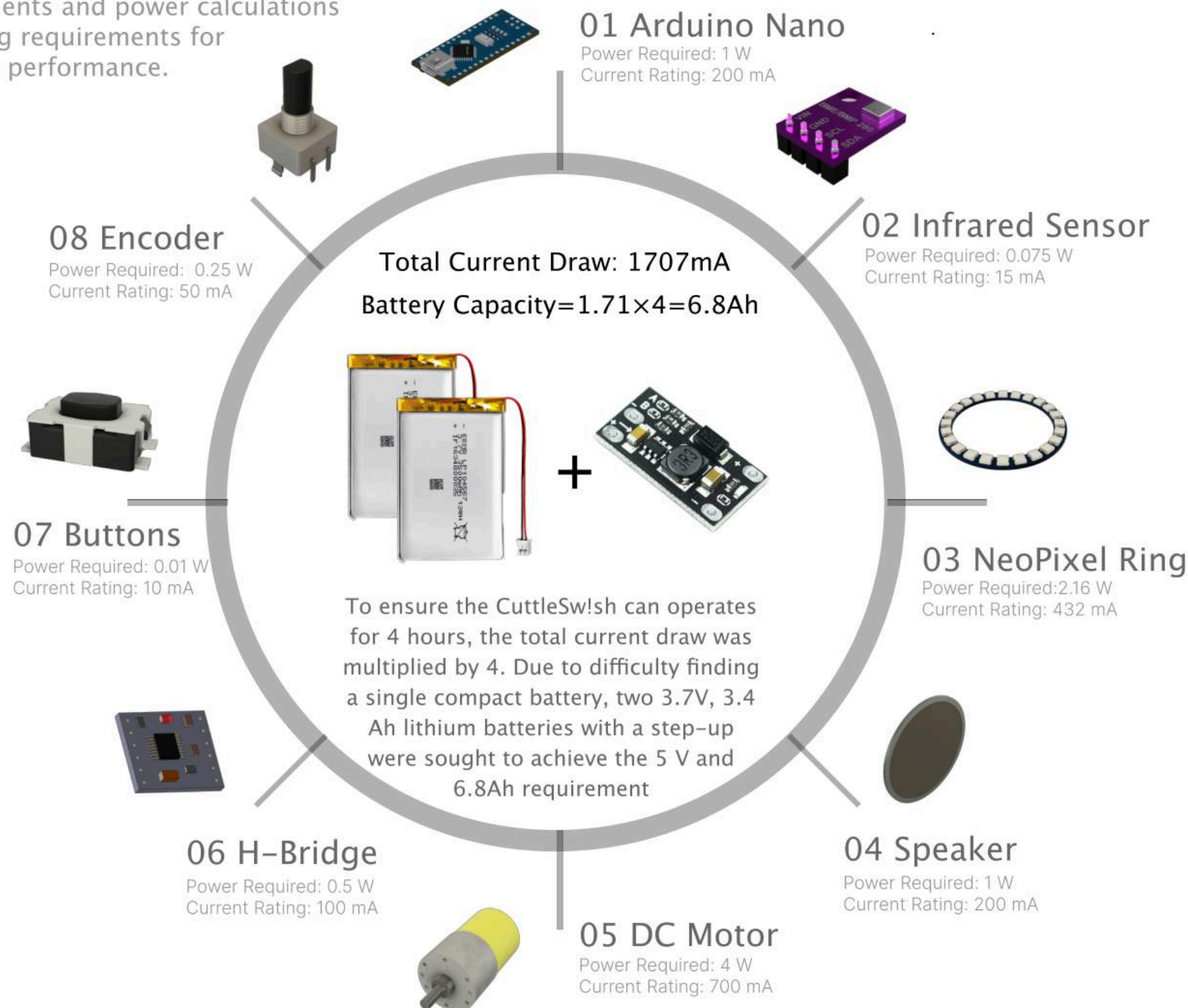


# SYSTEM ARCH!TECTURE

## POWER CALCULATIONS & COMPONENTS

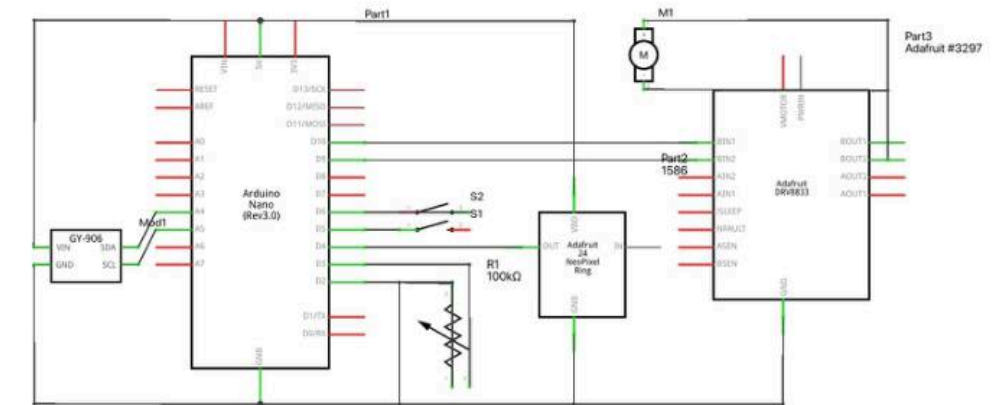
### Calculations

This diagram details the Cuttleswish components and power calculations outlining requirements for suitable performance.



### Architecture

The schematic and pseudo code illustrate the detailed circuitry and logic controlling the Cuttleswish.



System Schematic

```
SETUP {  
  init components, NeoPixel, MLX90614;  
}  
  
LOOP {  
  checkButton();  
  if power button pressed { togglePower(); }  
  
  if alarm {  
    if alarm duration > 5s { shutdown(); }  
    else { playAlarm(); }  
    RETURN;  
  }  
  
  if timer { updateTimer(); }  
  
  if active AND debounce delay {  
    if temperature mode { updateTempPixels(); }  
    else { handleEncoder(); }  
    lastUpdate = millis();  
  }  
}  
  
// Functions  
FUNCTION togglePower {  
  if active { shutdown(); }  
  else { turnOn(); colorIndex = 0; playMelody(powerOnMelody); updatePixels(); }  
}  
  
FUNCTION handleEncoder {  
  tmpdata = readEncoder();  
  if tmpdata {  
    if motor speed mode { blueCounter = tmpdata; updateMotor(); }  
    if timer mode { greenCounter = tmpdata; updateTimerPixels(); }  
  }  
}  
  
FUNCTION readEncoder { RETURN encoder change; }  
  
FUNCTION updatePixels {  
  if temperature mode { updateTempPixels(); }  
  else if motor speed mode { updateMotorPixels(); }  
  else if timer mode { updateTimerPixels(); }  
}  
  
FUNCTION updateTempPixels { temp = readTemp(); setPixels(red, temp); }  
  
FUNCTION updateMotorPixels { setPixels(blue, blueCounter); }  
  
FUNCTION updateTimerPixels { setPixels(green, greenCounter); }  
  
FUNCTION updateTimer {  
  elapsed = millis() - timerStart;  
  remaining = greenCounter - elapsed / 60000;  
  if remaining < 0 { shutdown(); playAlarm(); }  
  if timer mode { updateTimerPixels(); }  
}  
  
FUNCTION updateMotor {  
  if timer { pwm = map(blueCounter, 0, 1000, 0, 255); analogWrite(MOTOR_PDN_IN1, pwm); digitalWrite(MOTOR_PDN_IN2, LOW); }  
  else { stopMotor(); }  
}  
  
FUNCTION playMelody(melody) {  
  for each note { tone(note); delay(); noTone(); }  
}  
  
FUNCTION playAlarm {  
  for each note in alarmMelody { tone(note); delay(); }  
}  
  
FUNCTION clearPixels { for each pixel { pixels.setPixelColor(0); } pixels.show(); }  
  
FUNCTION shutdown {  
  active = false; timer = false; alarm = false;  
  noTone(12); playMelody(shutdownMelody); clearPixels(); stopMotor();  
}  
  
FUNCTION checkButton {  
  if button1 pressed { colorIndex++; if colorIndex > 2 { colorIndex = 0; } playMelody(buttonMelody); updatePixels(); }  
  if button2 long press AND timer mode AND greenCounter > 0 AND timer {  
    timer = true; timerStart = millis(); playMelody(timerStartMelody); updateMotor(blueCounter);  
  }  
}
```

Pseudo Code



UK & EU

Peleg Design products in the UK conform to both UK and EU requirements.



The UKCA marking certifies that our product is in compliance with UK safety regulations, meeting health, safety, and environmental standards.



The CE marking ensures adherence to EU safety, health, and environmental regulations allowing our product to be freely traded in the EU

Declaration

As part of ensuring the highest quality and safety, the Cuttleswish™ must include a declaration of conformity.

These document detail the manufacturer's name, address, and relevant product information, confirming that the product meets all applicable UKCA and CE standards. This documentation can be found in the user guide.



Implementation

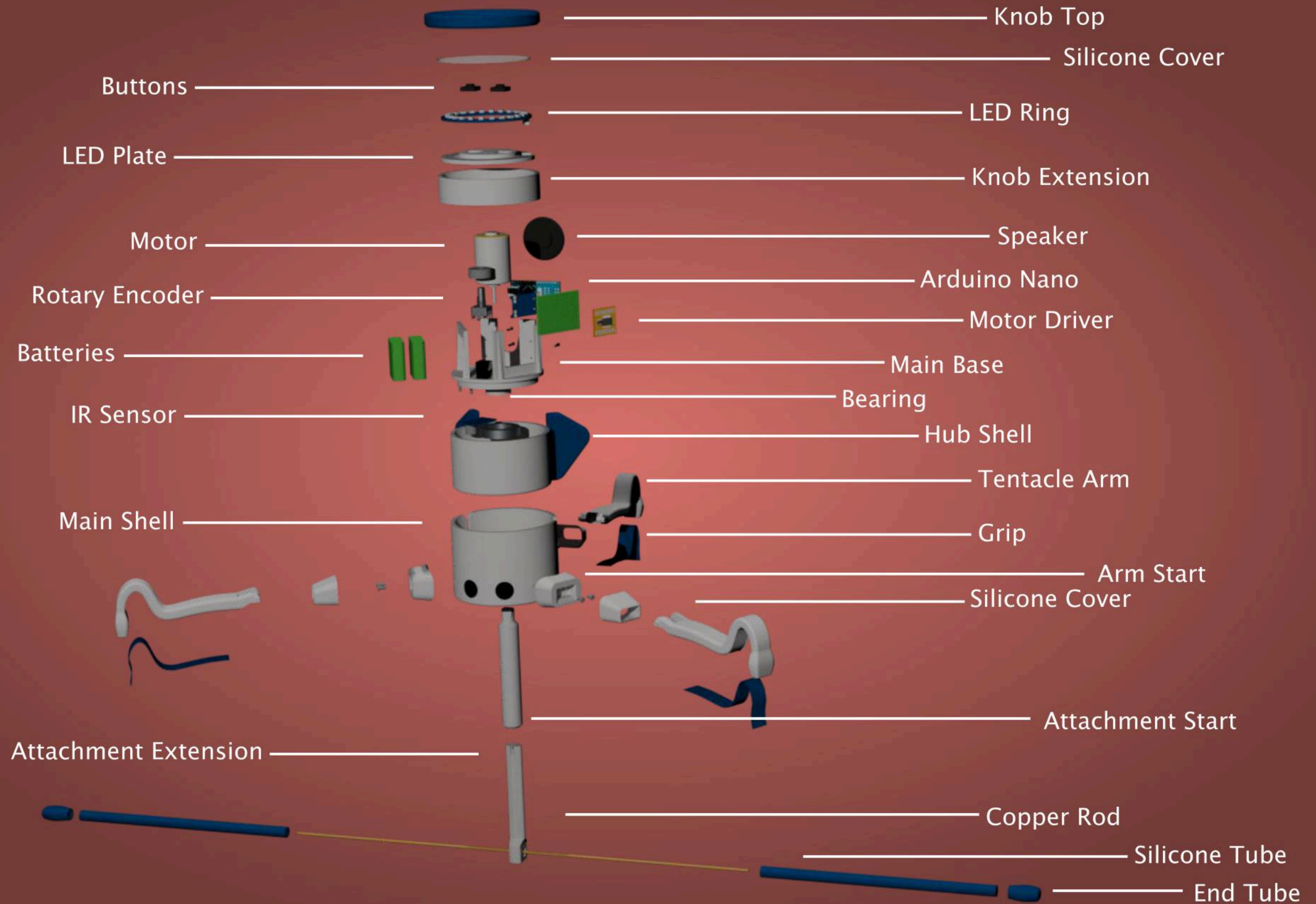
Known for its stylish and sleek finish, KitchenAid produces high-quality kitchen appliances that are both functional and visually appealing.





# EXPLODED VIEW

VISUAL RENDER





# PRODUCT LABELLING

## NAME & MARKINGS

### CUTTLESW!SH

Taking inspiration from Peleg Design's witty and humorous product names, we brainstormed Cuttlesw!sh, a cuttlefish which 'swishes' your food.

#### UKCA & CE

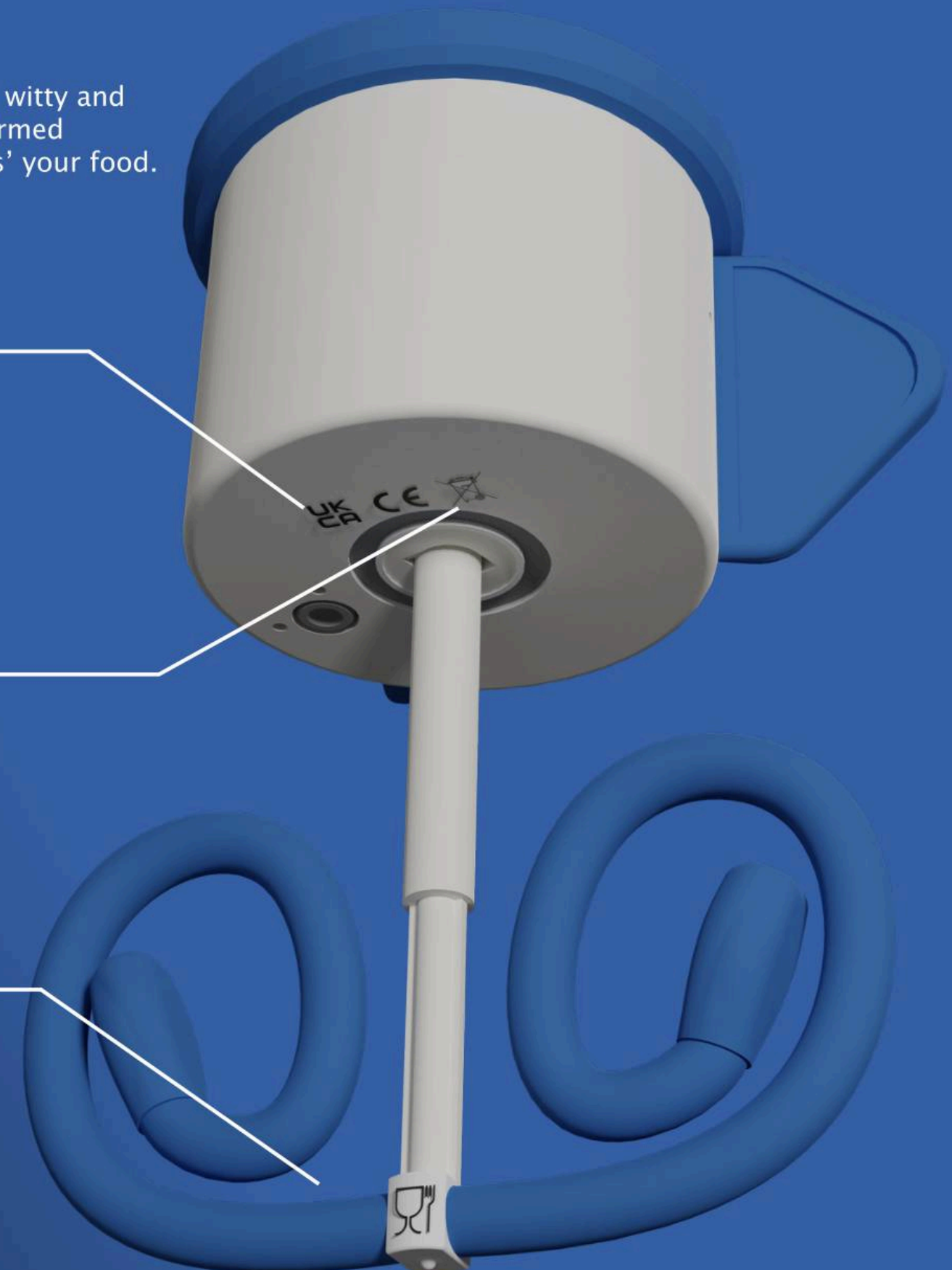
The UKCA (UK Conformity Assessed) and CE marks are mandatory labels that must be affixed to the product. These marks indicate compliance with relevant UK and EU regulations, ensuring the product meets safety, health, and environmental standards. [7]

#### WEEE

The Waste Electrical and Electronic Equipment (WEEE) directive focuses on reducing the environmental impact of electronic waste. Users are encouraged to recycle their Cuttleswish responsibly at end of life, opposed to disposing in household waste.

#### Food Contact Materials

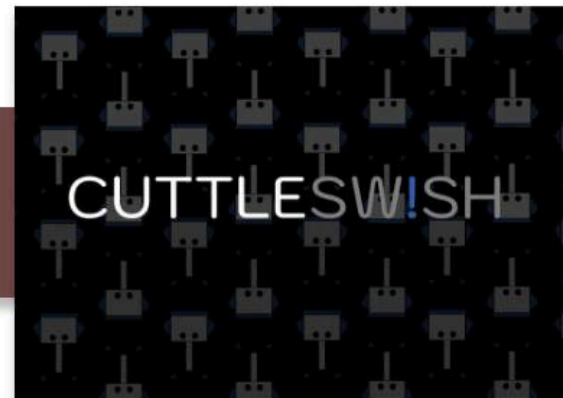
The food-safe symbol signifies that the materials used in manufacture are safe for use with food, adhering to regulations that limit harmful substances.





# USER GUIDE DESIGN

## INSTRUCTION MANUAL



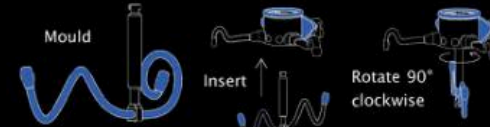
### Contents

|                           |   |
|---------------------------|---|
| Instructions              | 1 |
| Attachment Shapes         | 4 |
| Safety                    | 5 |
| Maintenance               | 6 |
| Support                   | 6 |
| Declaration of Conformity | 7 |

### Instructions

#### 1 Mould your Attachment

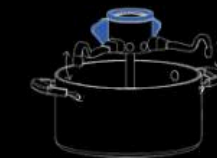
Depending on your aim for mixing we recommend a series of different popular shapes (See page X for diagrams).



Once your attachment is moulded insert and twist the attachment to lock into the central hub.

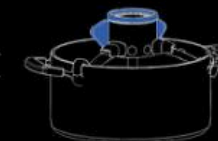
1

#### 2 Attach Cuttleswish™ to the pot



Angle the arms of your Cuttleswish™ to fit the size of your pot. The arms will allow pots of 19cm to 29cm.

When you attach the Cuttleswish™ the shaft of the attachment will retract to fit the your pot depth.



2

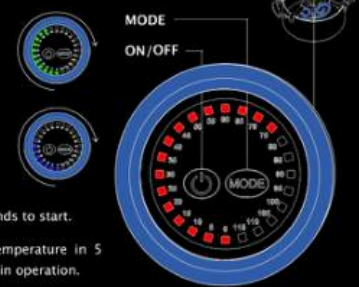
### 3 Controls

Press MODE and twist the knob clockwise to set the time in 5 minute increments.

Press MODE and twist the knob clockwise to set the speed.

Hold MODE for 2 seconds to start.

The ring will show temperature in 5 degree intervals when in operation.

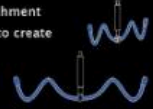


3

### Attachment Shapes

#### Wave: for wave-like stirring

Bend the attachment up and down to create waves.



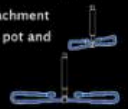
#### Butterfly: for fluttery whisking

Bend the attachment up and around proportion to your pot size



#### Fin: for sweeping and scraping

Straighten the attachment to the edge of the pot and bend round.



#### Helix: for helical mixing

Twist each attachment in opposite direction it in a corkscrew motion.



4

### Safety

#### 1 Cleaning

Once the hub is removed all parts can be washed and are dishwasher safe.



#### 2 Storage and Disposal

Store your Cuttleswish™ in a cool dry place when not in use. Dispose of the product in accordance with the Waste Electrical and Electronic Equipment (WEEE) Regulations. Do not dispose of in household waste.



#### 3 Attachment

Keep hands and utensils away from the attachment while operating to prevent injury or device damage.



5

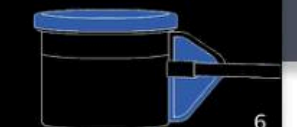
### Safety

#### 4 Charging

Hold both fins and lift the central hub from the outer shell.



After removal, wipe the central hub with a wet cloth if dirty. Once dry, charge your Cuttleswish™ for X hours for a full charge.



6

### Maintenance

Clean the blades and hub with a damp cloth and ensure that no water enters the motor housing. Avoid using harsh chemicals or abrasive materials. After cleaning and drying, reassemble by aligning the fins and pushing the central hub back into place.



### Support

For more information and further help please contact us at [shop@pelegdesign.com](mailto:shop@pelegdesign.com). For an enlarged version of the manual please scan this QR code. If you experience any issues with the Cuttleswish™ or need replacement parts, our support team is here to assist you.



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### UK Declaration of Conformity

This declaration of conformity is issued under the sole responsibility of Peleg Design. The product satisfies the provision for CE marking according to the following UK regulations:

The Restriction of the Use of Certain Hazardous Substances in Electrical and Electronic Equipment Regulations 2012 (SI 2012/2020)  
The Waste Electrical and Electronic Equipment Regulations 2013 (SI 2013/2113)  
The Ecodesign for Energy-Related Products Regulations 2012 (SI 2012/2617)  
The Electromagnetic Compatibility Regulations 2012 (SI 2012/1992)  
The Batteries and Accumulators (Placing on the Market) Regulations 2008 (SI 2008/1644)  
The Materials and Articles in Contact with Food (England) Regulations 2012 (SI 2012/2618)

Type of equipment: Automatic Stirrer  
Brand name or trade mark: Cuttleswish™  
Type designation: 1000 (e.g. Model No.)  
Software: Firmware: N/A  
Manufacturer: Peleg Design  
Address: Sharncliffe Str. 20, 71395 Osnabrück, Deutschland  
Telephone No.: 0049 - 07033 - 46668-28  
UK Representative: N/A

The following dedicated standards or technical specifications, which comply with good engineering practice in safety matters in force within the UK, have been applied:

RoHS: EN 50581  
WEEE: EN 50625  
EMC: EN 55032  
EN 61000-3-2  
EN 61000-3-3  
EcoDesign: EN 50564  
EN 60960-1  
Food Contact Materials Regulation: EN 1186



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### EU Declaration of Conformity

This declaration of conformity is issued under the sole responsibility of Peleg Design. The product satisfies the provision for CE marking according to the following directives:

RoHS Directive 2011/65/EU  
Electromagnetic Compatibility Directive 2014/53/EU (EMC 2014/53/EU)  
Waste Electrical and Electronic Equipment Directive 2012/19/EU (WEEE 2012/19/EU)  
Eco-design Directive 2009/125/EC  
Battery Directive 2006/66/EC  
Regulation (EC) No 1935/2004 (Food Contact Materials)

Type of equipment: Automatic Stirrer  
Brand name or trade mark: Cuttleswish™  
Type designation: 1000 (e.g. Model No.)  
Software: Firmware: N/A  
Manufacturer: Peleg Design  
Address: Sharncliffe Str. 20, 71395 Osnabrück, Deutschland  
Telephone No.: 0049 - 07033 - 46668-28  
UK Representative: N/A

The following harmonised European standards or technical specifications, which comply with good engineering practice in safety matters in force within the EEA, have been applied:

RoHS: EN 50581  
WEEE: EN 50625  
EMC: EN 55032  
EN 61000-3-2  
EN 61000-3-3  
EcoDesign: EN 50564  
EN 60960-1  
Food Contact Materials Regulation: EN 1186



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## Instructions

p. 1-3

Detailed step-by-step guide for assembling and using Cuttleswish with descriptions and infographics, ensuring user-friendly operation and ease of use.

## Attachment Shapes

p. 4

Overview of various mouldable silicone attachments, highlighting their adaptability to different pot shapes and sizes for efficient mixing.

## Safety

p. 5-6

Comprehensive safety guidelines and warnings to ensure safe operation and prevent accidents, crucial for user protection.

## Maintenance and Support

p. 7

Tips for maintaining Cuttleswish and support information, ensuring longevity and provide customer assistance, including a QR code for a digital enlarged guide.



# PACKAGING DESIGN

EXPLODED VIEW

The Cuttleswish packaging design facilitates an engaging unboxing experience, highlighting the product's quality and thoughtful details. Each layer, from the outer box to the base compartment, is designed for ease of use, protection, and visual appeal. This design not only protects the product but also delights the user from the first moment they open the box.

## Labelling

Packaging is clearly labeled with the product name, description, company logo, and regulatory symbols: UKCA, CE, WEEE and FCM. (Package Disposal symbol are featured on the underside.)

The QR code featured provides easy access to the user manual and support information.

## Base Compartment

The base compartment is made from durable cardboard and securely stores the USB charging cable, user guide and mouldable attachment.

It features moulded slots for organisation and protection, ensuring stability and ease of access during transit. Manufactured using precise die-cutting, it combines functionality with sturdiness.



## Outer Box

Made of high-quality non-corrugated cardboard. This features branded graphics and instructions, created using die-cutting and printing processes.

## Protective Foam Layer

Constructed from HDPE foam, this insert secures the Cuttleswish during transport. It is precisely moulded and cut for a snug fit, ensuring protection from damage.

## Branding Wrap

Made from printed tissue paper, designed with a tessellation of Cuttleswish icons. It is produced through printing and die-cutting to fit neatly around the packaging.





CUTTLESWISH

Cuttleswish™  
Automatic Stirrer for Elderly

PELEGDESIGN®

Step 1

Attachment  
The three arms snap to fit pot diameters 19cm and 29cm to cater to your biggest and smallest pots.



Step 2

Attach Cuttleswish™ to your pot.

The shaft height will adjust when attaching your Cuttleswish™ to ensure the attachment is the perfect height for your pot.



Step 3

Set time, speed and go!

Twist the knob when on green to set the time. Each LED represents 5 minutes of stirring.

Twist the knob when on blue to set the speed. Then sit back, relax and let Cuttleswish™ do the heavy lifting.





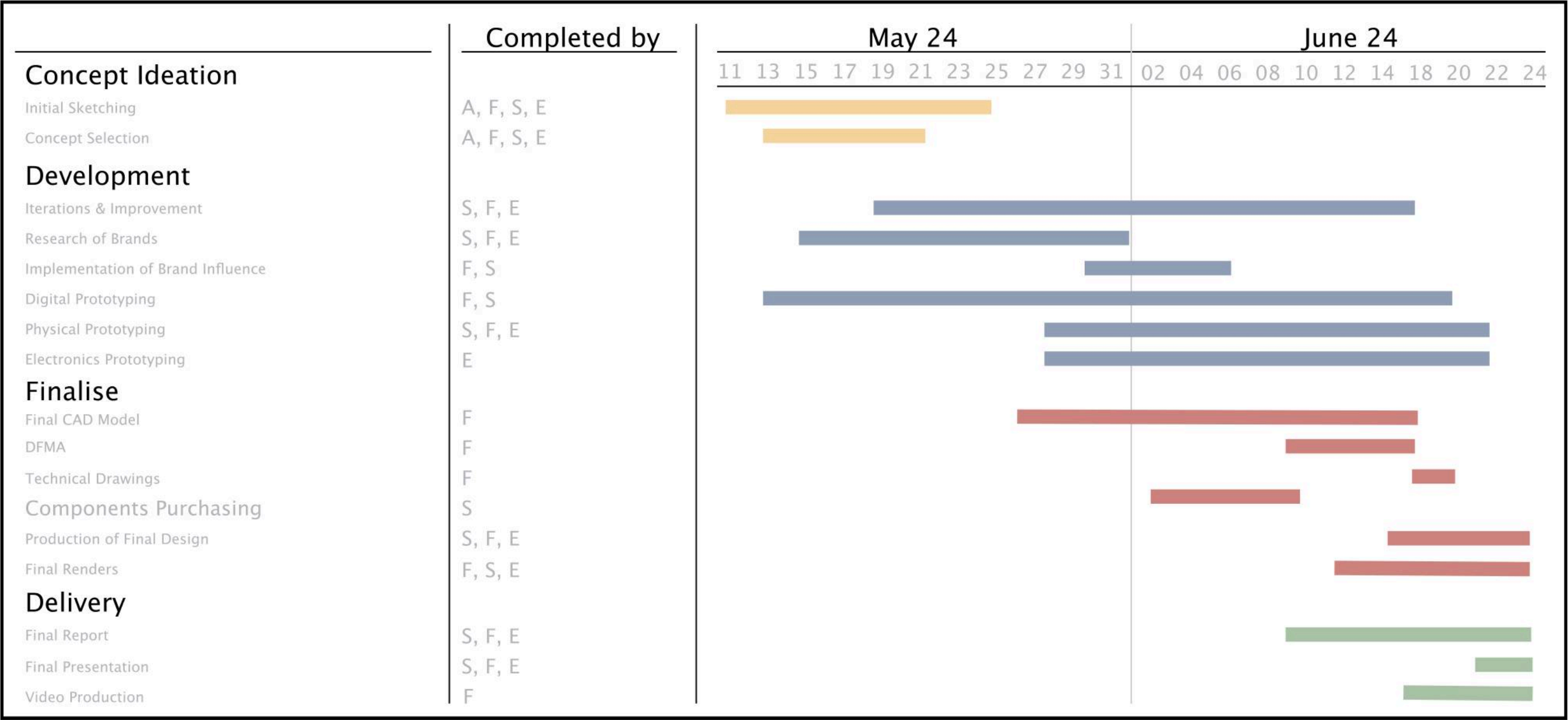








Gantt Chart

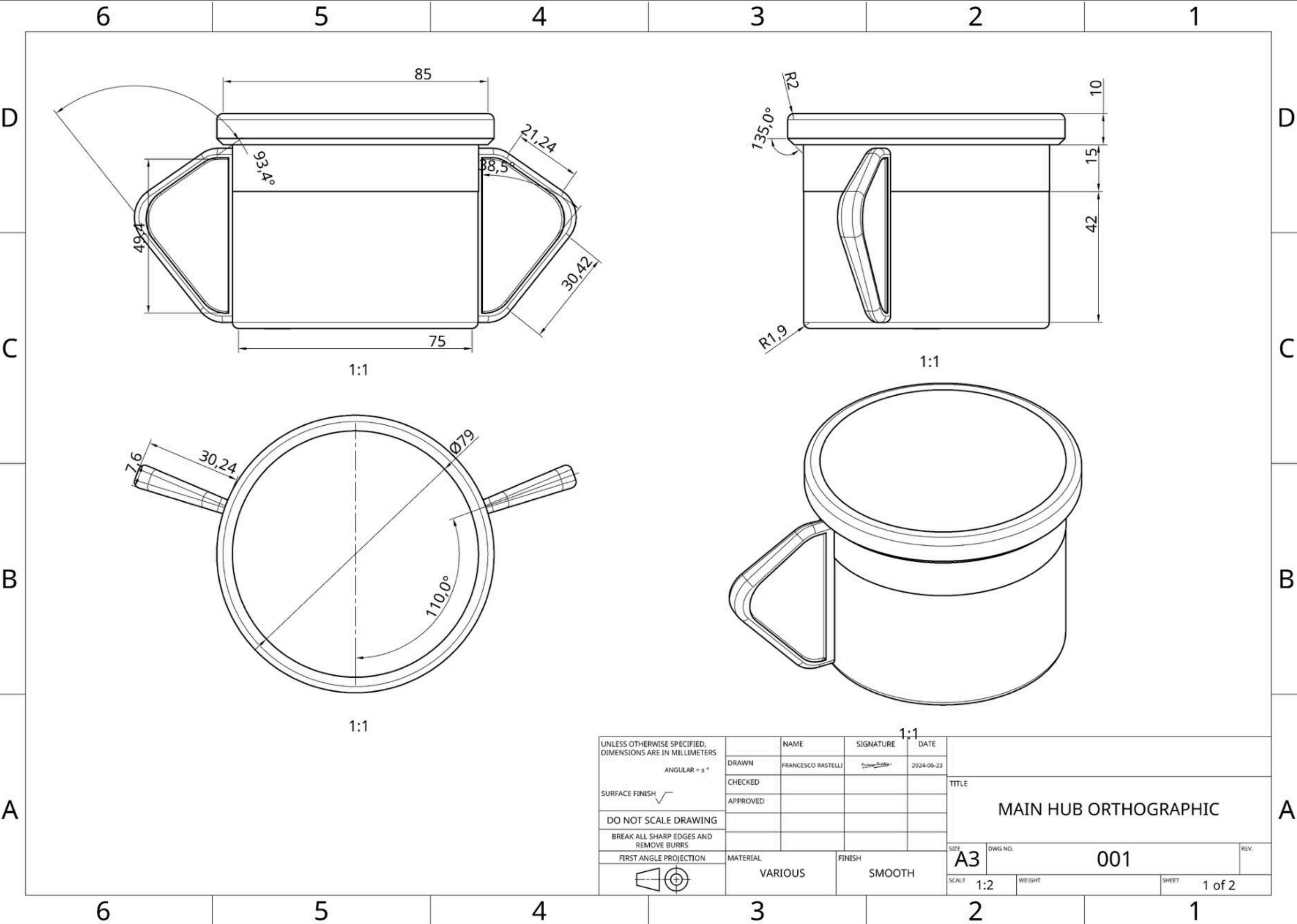


Weekly meetings proved vital in the early stages for our divergent stages of concept development. These meetings ensure that we consistently on track with weekly tasks.

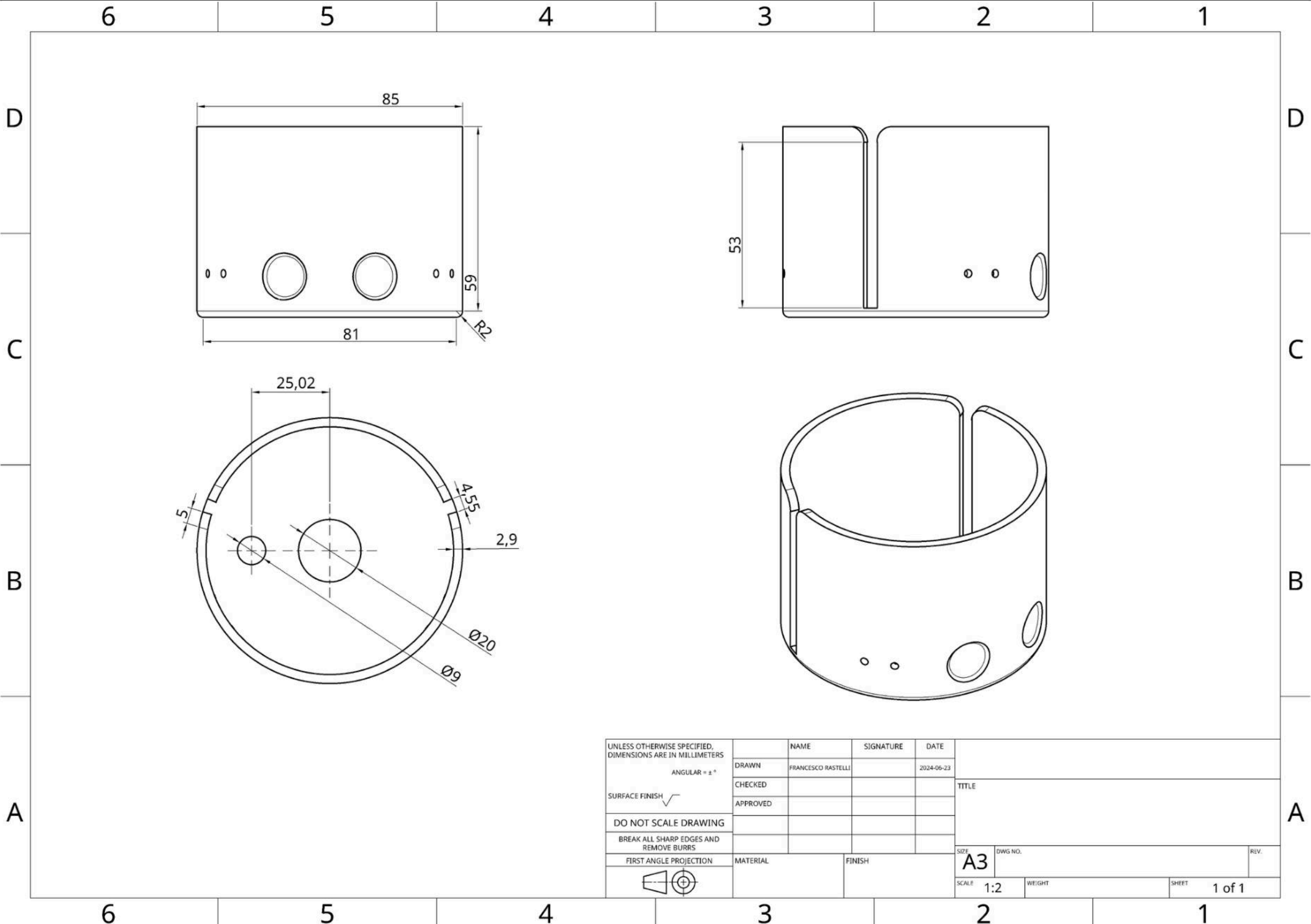
Conducting initial ideation together followed ensured we all had equal input into the final concept. From this point onwards delegating tasks proved the most efficient way for steady progress.

Despite facing multiple challenges and obstacles with team dynamics. With perseverance and teamwork we were able to complete deliverables to the best of our ability.











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